

Shiqi Ding (She/Her/Hers)

Ph.D. Candidate, Interdisciplinary Urban Design and Planning, University of Washington, Seattle, WA

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Education

2023 – present	University of Washington Ph.D. in Interdisciplinary Urban Design & Planning <i>Dissertation:</i> Urban Dynamics and Transportation Electricity Demand: Spatial and Temporal Forecasting Methods for Infrastructure Planning <i>Committee:</i> Lingzi Wu (Chair), Qing Shen, Don MacKenzie, Narjes Abbasabadi (GSR) Ph.D. Track in Statistics, Center for Statistics and the Social Sciences (CSSS), 2026 Graduate Certificate in Clean Energy Science, Engineering, & Society, Clean Energy Institute (CEI), 2025	Seattle, WA
2022 – 2023	Clemson University Ph.D. student in Civil Engineering	Clemson, SC
2016 – 2018	Politecnico di Milano M.S. in Architecture	Milan, Italy
2011 – 2016	Wuhan University Bachelor of Architecture	Wuhan, China

Research Interests

Topics: Transportation electrification; Urban energy demand and grid impacts; EV charging infrastructure; Resilience; Weather and climate impacts on mobility and charging demand

Methods: Spatial statistics; Bayesian methods; Time-series models; Spatial-temporal forecasting; Probabilistic machine learning and deep learning models; Uncertainty quantification

Publications

Peer-Reviewed Journal Articles:

- [1]. **Ding, S., & Wu, L.,** (2025). Disparities in electric vehicle charging infrastructure distribution: A socio-spatial clustering study in King County, Washington. *Sustainable Cities and Society*, 121, 106193. <https://doi.org/10.1016/j.scs.2025.106193>.
- [2]. Dong, B., **Ding, S.,** Wu, L., & Li, X., (2025). Short-term natural disaster impacts on transportation infrastructure: A Systematic Review. *Natural Hazards*, 121, 17321-17362. <https://doi.org/10.1007/s11069-025-07504-9>.
- [3]. **Ding, S.,** Tang, C., Li, D. & Lopes, S., (2023). Enhancing energy efficiency and thermal satisfaction in office buildings: A comprehensive evaluation of space match strategy. *Energy and Buildings*, 298, p.113526. <https://doi.org/10.1016/j.enbuild.2023.113526>.

Peer-Reviewed Conference Papers:

- [1]. **Ding, S., & Wu, L.,** (2026). Temperature Effects on Hourly Electric Vehicle Charging Demand Across Temporal Contexts. In *ASCE International Conference on Computing in Civil Engineering (i3CE)*.
- [2]. **Ding, S., & Wu, L.,** (2025). Analyzing Public Electric Vehicle Charging Distribution and Urban Socio-Spatial Characteristics Using Geographically Weighted Machine Learning. In *ASCE*

International Conference on Computing in Civil Engineering (i3CE), (pp. 263-272), <https://doi.org/10.1061/9780784486443.030>.

- [3]. **Ding, S.**, Tang, C., Lopes, S. & Li, D., (2024). Assessing energy performance and thermal satisfaction of flexible space usage in office buildings. In *Construction Research Congress (CRC)*, (pp. 316-326), <https://doi.org/10.1061/9780784485279.033>.

Peer-Reviewed Conference Presentations & Abstracts:

- [1]. **Ding, S.**, & Wu, L., (2027). Scale, Style, and Speed: Characterizing Utilization Growth Trajectories at Curbside Electric Vehicle Charging Sites. In *Transportation Research Board (TRB) Annual Meeting, Jan 2027*. [Accepted]
- [2]. **Ding, S.**, Wu, L., & Shen, Q. (2026). A Cross-Classified Multilevel Analysis of Workers' Commuting Adaptive Capacity. In *Association of Collegiate Schools of Planning (ACSP) Annual Conference, Oct 2026*.
- [3]. **Ding, S.**, & Wu, L., (2025). Spatial Patterns of Electric Vehicle Miles Traveled (EVMT) as A Proportion of Total Mobility. In *Association of Collegiate Schools of Planning (ACSP) Annual Conference, Oct 2025*.

Manuscripts Under Review and in Preparation:

- [1]. **Ding, S.**, & Wu, L. (2026). Estimating Electric Vehicle Miles Traveled: A Bayesian Hierarchical Spatial Zero-Inflated Beta Model. Preprint: <https://doi.org/10.2139/ssrn.6161434>.
- [2]. **Ding, S.**, & Wu, L. Scale, Style, and Speed: Characterizing Utilization Growth Trajectories at Curbside Electric Vehicle Charging Sites. [In preparation]
- [3]. **Ding, S.**, Chen, L., Tang, C., Chen, K., & Wu, L. The Effect of Perceived Weather on Short-Term EV Charging Load Forecasting. [In preparation]

Research Experience

- 2023 – present **Graduate Research Assistant** University of Washington
Department of Urban Design and Planning and Department of Construction Management
(Advisor: Prof. Lingzi Wu)
– Developed a Bayesian hierarchical spatial model to estimate census-tract EV miles traveled from sparse household travel survey data.
– Analyzed utilization growth and forecast daily charging load for public curbside EV chargers in Seattle.
– Modeled temperature effects on hourly EV charging demand across temporal contexts.
- 2024 – 2025 **Science Policy Analyst** University of Washington
Advanced Experience Program (AXP), Clean Energy Institute
– Co-wrote technical policy briefs and white papers with the Washington State Academy of Sciences (WSAS).
- 2022 – 2023 **Graduate Research Assistant** Clemson University
College of Engineering, Computing and Applied Sciences

Grants and Funding

- [1]. **Characterizing Charger Utilization Growth Trajectories in Public EV Charging Networks**
UW College of Built Environments, 2026 Summer Student Funding, \$3,250. [Funded]
Role: Co-PI; co-wrote the proposal. PI: Prof. Lingzi Wu.

- [2]. **An Integrated Human-System Framework for Behavioral Resilience in Multimodal Transportation under Weather Disruptions**
Pacific Northwest Region 10 University Transportation Center (PacTrans), 2026–2028 Small Project, \$99,668. **[Submitted, not funded]**
Role: Contributed to proposal writing. PI: Prof. Lingzi Wu; Co-PI: Prof. Qing Shen.

Honors and Awards

- 2026 **Outstanding Ph.D. Student Award**
Department of Urban Design and Planning, UW
- 2025 **Most Inspiring and Best Use of Visuals Awards**
CEI Graduate Fellow Lightning Talk, Clean Energy Institute, UW
- 2024 – 2025 **CEI Graduate Fellowship (two quarters of research assistantship)**
Clean Energy Institute, UW

Conference Presentations and Posters

- 01/2027 Scale, Style, and Speed: Characterizing Utilization Growth Trajectories at Curbside Electric Vehicle Charging Sites. *TRB Annual Meeting, Washington, DC.* [Accepted]
- 10/2026 Commuting Adaptive Capacity: A Cross-Classified Multilevel Analysis of Workers' Weekly Commute Patterns. *ACSP Annual Conference, Pittsburgh, PA.*
- 06/2026 Temperature Effects on Hourly Electric Vehicle Charging Demand Across Temporal Contexts. *i3CE, Songdo, South Korea (Remotely).*
- 04/2026 Spatial Patterns of Electric Vehicle Miles Traveled (EVMT). *CBE Research Showcase and Symposium, Seattle, WA.* [Poster]
- 10/2025 Spatial Patterns of Electric Vehicle Miles Traveled (EVMT) as a Proportion of Total Mobility. *ACSP Annual Conference, Minneapolis, MN.*

Invited Talks and Seminars:

- 10/2025 Estimating EV Usage Given Sparse Survey Data: A Bayesian Hierarchical Spatial Zero-Inflated Beta Modeling Framework. *Urban Design & Planning Ph.D. Seminar, UW, Seattle, WA.*
- 08/2025 Planning for Sustainable Urban Mobility. *Washington Transportation High School Summer Camp, Seattle, WA.*

Teaching Experience

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| Autumn 2025 | Instructor of Record
R E 497/597: Real Estate Data Modeling | Runstad Department of Real Estate, UW |
| Autumn 2024 | Instructor of Record
R E 497/597: Real Estate Data Modeling | Runstad Department of Real Estate, UW |
| Spring 2024 | Teaching Assistant
R E 497/597: Real Estate Data Modeling | Runstad Department of Real Estate, UW |
| Winter 2024 | Teaching Assistant
R E 397: Introduction to Real Estate Data Modeling | Runstad Department of Real Estate, UW |

Professional Service

Journals

Reviewer	Building and Environment	2 articles	2025
	Journal of Building Engineering	1 article	2025
	Journal of Construction Engineering and Management	1 article	2025

Conferences

Reviewer	International Symposium on Automation and Robotics in Construction (ISARC)	7 articles	2024-2026
	Transportation Research Board Annual Meeting (TRB)	5 articles	2025-2026

Outreach

2024 – 2025	Student Project Mentor and Invited Speaker	Seattle, WA
	Washington Transportation High School Summer Camp (sponsored by WSDOT and PacTrans)	
05/2025	Clean Energy Outreach Volunteer	Renton and Shoreline, WA
	Middle school solar-car activities and STEM night exhibitor	

Professional Experience

2020 – 2022	Architect	Guangzhou, China
	<i>XAA Janson Xian Architects</i>	
2018 – 2019	Design Associate	Shanghai, China
	<i>WeWork Inc.</i>	

Technical Skills

Programming: R, Python **Modeling:** Stan/brms, INLA, PyTorch **GIS:** ArcGIS Pro, QGIS

Professional Certification and Memberships

2024 – present	Association of Collegiate Schools of Planning (ACSP), Member	
2023 – present	American Society of Civil Engineers (ASCE), Member	
2019	LEED AP BD+C	Green Business Certification Inc.